Demo: Research Agent in n8n Cloud

Connect an n8n AI agent to your own MarketIntel MCP server running on your laptop

Objective: Build an AI agent in n8n Cloud that uses the MarketIntel MCP server as its tool, then ask it for a market-research report while your server terminal shows every call arriving.

Time needed: about 30 minutes to build the first time, then about 10 minutes to run the demo.

What you will have at the end: a working chat window in n8n where an agent (powered by Google Gemini, with memory) calls your MarketIntel tools through a public tunnel.

1. How it works

n8n Cloud runs on n8n's servers, so it cannot see programs on your laptop. The MarketIntel server therefore needs a public HTTPS address, which the ngrok tunnel provides. The agent in n8n reads the server's tool list and decides which tools to call.

Inside the n8n workflow you will create four pieces:

- **Chat Trigger:** the chat window where you type questions.
- **AI Agent (named Research Agent):** the brain that decides what to do.
- **Google Gemini Chat Model + Simple Memory:** the language model and its short-term memory.
- **MCP Client Tool:** the connector that reaches your MarketIntel server.

2. Before you start

n8n Cloud account	Hosts the workflow and chat window	Your existing account (check that your trial or plan still has executions left)
Google account	To create a free Gemini API key	aistudio.google.com
MarketIntel project on your laptop	The MCP server the agent will use	From the first guide: server.py, test_client.py, .env
ngrok account and authtoken	Gives the server a public HTTPS address	From the first guide, Steps 6 and 7 (dashboard.ngrok.com)
Tavily API key (optional)	Live web results. Without it the server uses sample data for NVIDIA, AMD and Intel	tavily.com
Two terminal windows	One for the server, one for ngrok	Terminal on your Mac

Pre-flight checklist. Tick these off before Part A:

- You can log in to your n8n Cloud workspace.
- The MarketIntel folder runs: uv run server.py prints the banner (you did this in the first guide).
- ngrok is installed and your authtoken is saved (ngrok config add-authtoken ...).

- You have a Google account you can sign in with.

Starting from scratch? If you have not built the MarketIntel server or ngrok yet, complete Steps 1 to 7 of the first guide (MarketIntel MCP Claude Demo Guide) and come back here.

3. Part A: Start the MarketIntel server and the tunnel

Do this first. n8n must be able to reach a running server when you connect the tool.

A1. Start the server

- Open a terminal window (Terminal 1).
- Go to the project folder and start the server (use your own folder path):

```
cd marketintel_demo
uv run server.py
```

- Read the banner. The **Mode** line tells you what data the demo will use:
- LIVE (Tavily): real web results (best for the demo).
- DEMO (canned data...): sample data only for NVIDIA, AMD and Intel. Fine as a fallback.
- Leave Terminal 1 open. This is your live log window.

Using the plain server? If you use server_plain.py instead, start it with TRANSPORT=http uv run server_plain.py. It works the same way but prints no logs, so you lose the "who called what" view. server.py is recommended for this demo.

A2. Test the server locally

- Open a second terminal (Terminal 2) in the same project folder.
- Run:

```
curl http://localhost:8000/health
```

- You should see JSON containing "status":"ok".

A3. Start the ngrok tunnel

- In Terminal 2, start the tunnel:

```
ngrok http 8000
```

- If ngrok complains about authentication (error ERR_NGROK_4018), save your authtoken once and run the command again. The token is on dashboard.ngrok.com/get-started/your-authtoken:

```
ngrok config add-authtoken <YOUR_TOKEN>
```

- Find the **Forwarding** line and copy the https address, for example https://a1b2-203-0-113-9.ngrok-free.app.
- Add **/mcp** to the end. This full address is your MCP URL. Write it down or keep it in a note:

```
https://a1b2-203-0-113-9.ngrok-free.app/mcp
```

- Verify it from a browser: open your ngrok address plus /health. If ngrok shows a warning page, click **Visit Site** (this page appears only in browsers). You should see the same JSON as in A2.

Checkpoint: Terminal 1 (server) and Terminal 2 (ngrok) are both running, and the /health address works in your browser. Do not close either window.

4. Part B: Get a free Gemini API key

The agent needs a language model. This guide uses Google Gemini because it has a free tier (with usage limits).

- Open https://aistudio.google.com/app/apikey in your browser and sign in with your Google account. Accept the terms if asked.
- Click **Create API key**.
- Choose **Create API key in new project** (or pick an existing project).
- Copy the key that appears. Keep it private, like a password. You will paste it into n8n in Part C (Step C6).

5. Part C: Build the agent in n8n Cloud

Menu names in n8n change from time to time. If a label differs slightly, look for the closest match.

C1. Open n8n and create a workflow

- Log in to your n8n Cloud workspace in the browser.
- On the Overview page, click **Create workflow** (top-right corner). A blank canvas opens.
- Click the workflow name at the top left (it says something like “My workflow”) and rename it to **Research Agent - MarketIntel**. Press Enter.

C2. Add the chat trigger

- On the canvas click **Add first step**.
- In the search box type **Chat Trigger** and select it. It may appear as “On chat message” or “When chat message received”.
- A settings panel opens. Leave everything as it is and click **Back to canvas** (top-left).

C3. Add the AI agent

- Click the **+** button at the right edge of the Chat Trigger node.
- Search for **AI Agent** and select it.
- When the agent’s panel opens, click **Back to canvas**. You now see the agent node with three small + connectors underneath: **Chat Model**, **Memory** and **Tool**.

C4. Add the Gemini chat model

- Click the **+** under **Chat Model**.
- Search for **Google Gemini Chat Model** and select it.
- In the **Credential** dropdown click **Create new credential**. A credentials window opens.
- Leave the **Host** as it is (the default Google address). Paste your Gemini key into the **API Key** field.

- Click **Save** (top-right of the credentials window). n8n usually tests the key and confirms it works.
- Close the credentials window with the **X**.
- In the **Model** field choose a **Flash** model from the list. Flash models are quicker and lighter on free-tier limits.
- Click **Back to canvas**.

C5. Add memory

- Click the **+** under **Memory**.
- Select **Simple Memory**. Leave its settings at the defaults.
- Click **Back to canvas**.

Memory lets the agent remember earlier messages, so you can ask follow-up questions in the demo.

C6. Add the MCP Client Tool (the MarketIntel connection)

This is the most important step. It connects the agent to your server.

- Click the **+** under **Tool**.
- Search for **MCP Client Tool** and select it.
- Fill in the node settings exactly like this:

Endpoint	Your ngrok MCP URL from Step A3, ending in /mcp (for example https://a1b2-203-0-113-9.ngrok-free.app/mcp). In some n8n versions this field is called “SSE Endpoint”; use the same URL.
Server Transport	HTTP Streamable
Authentication	None
Tools to Include	All

Do not skip Server Transport: the default is SSE, which is the older style. MarketIntel speaks HTTP Streamable. With the wrong transport the node connects to nothing, and the agent will see an empty tool list.

- Click **Back to canvas**.

C7. Rename the agent

- Double-click the **AI Agent** node to open it.
- Click the node name at the top of the panel (“AI Agent”), type **Research Agent** and press Enter.

C8. Add the system message

The system message tells the agent how to behave and which tools to use.

- With the Research Agent panel still open, scroll down and click **Add Option**.
- Choose **System Message**.
- Delete the default text and paste this message:

```
You are a helpful market research assistant. Always use the MarketIntel
```

tools (company_overview, list_competitors, product_portfolio, pricing_snapshot, recent_news_pulse, competitor_brief) to gather facts before you answer. Cite the source URLs the tools return. If the tools return no data, say so instead of guessing. Keep answers concise.

- Click **Back to canvas**.

C9. Save the workflow

- Click **Save** (top-right), or press Ctrl+S on Windows / Cmd+S on a Mac.

Checkpoint: Your canvas matches the picture in Section 1: Chat Trigger on the left feeding the Research Agent, with Google Gemini Chat Model, Simple Memory and MCP Client Tool attached underneath.

6. Part D: Test the connection

Do a quick test before anyone is watching.

- At the bottom of the canvas click **Open chat**. A chat panel appears.
- Type this message and press Enter:

Which tools do you have access to? List them with a one-line description.

- Expect an answer that lists the MarketIntel tools: company_overview, list_competitors, product_portfolio, pricing_snapshot, recent_news_pulse, competitor_brief (and server_status if you run server.py).
- Switch to Terminal 1. You should see new lines. They look like this (an illustrative example; your IP, host and client name will differ, and the calls come from n8n's servers, not your laptop):

```
[11:20:14.101] [HTTP  ] --> POST /mcp session=- ip=203.0.113.9 host=a1b2.ngrok-free.app ua='<sent by n8n>'
[11:20:14.103] [INIT  ] New MCP session: client='<name sent by n8n>' ip=203.0.113.9 ...
[11:20:14.420] [LIST  ] Client asked for tools -> returned 7: ['company_overview', ...]
```

Checkpoint: The agent lists the MarketIntel tools and Terminal 1 shows [HTTP], [INIT] and [LIST] lines. The connection works. If not, go to the Troubleshooting section.

7. Part E: Run the demo

Arrange your screen so the audience can see the n8n chat on one side and Terminal 1 on the other.

E1. Prompts to use

1	Which tools do you have access to? List them.	Terminal: [INIT] and [LIST]. Proves the agent discovered the server.
2	Give me a company overview of NVIDIA.	Terminal: [CALL] company_overview with its arguments, then [TAVILY] or [DEMO], then [DONE] with timing.

3	Who are AMD's main competitors in GPUs?	Terminal: [CALL] list_competitors with name=AMD, category=GPUs.
4	Create a market research report comparing NVIDIA and AMD. Cover: 1. Company overview and market position 2. Product portfolio (gaming GPUs, AI/data center chips, etc.) 3. Pricing and recent price trends 4. Key competitors (Intel, Qualcomm, etc.) and positioning 5. Strategic moves and future outlook. Keep it concise (under 300 words) and use credible sources.	Several [CALL] lines in a row. The agent chooses and chains tools by itself, then writes the report. This is the main moment of the demo.
5	Now add Intel to the comparison and tell me who has the lowest prices.	Shows memory: the agent remembers the earlier report and calls tools again for Intel.
6	Which MarketIntel tools did you use for that answer?	The agent explains its own tool use. The terminal log confirms it.

Tip: Run prompt 4 once before the audience arrives. If the free Gemini limit is hit mid-demo, wait a minute and continue. Using NVIDIA, AMD or Intel keeps demo mode useful if Tavily is not set up.

E2. Where to point during the demo

- **In n8n:** open the Research Agent node after a run and look at its output, or open the logs under the chat. Each tool call the agent made is shown there.
- **In Terminal 1:** follow the lines:
  - [CALL] shows which tool and which arguments the agent chose.
  - [TAVILY] or [DEMO] shows where the data came from.
  - [DONE] shows how long it took.
- You may see a new [INIT] line for each message you send. That is normal; the agent connects when it starts working.

E3. Suggested presenter flow (about 10 minutes)

1 min	Show the canvas picture and the two terminals.	"This is an AI agent. It has a brain, a memory, and one tool connection. The tools live on my laptop, not in n8n."
2 min	Run prompt 1.	"The agent asks the server what it can do. Watch the log: the server just saw that request."
3 min	Run prompts 2 and 3.	"The agent decides which tool to call and what to send. I never told it."
3 min	Run prompt 4 (the report).	"It calls several tools, combines the results, and writes the report with sources."
1 min	Run prompt 5 (memory), then wrap up.	"Same server works with Claude too. Build the server once, plug it into any client."

E4. Optional: show control over the tools

- Open the **MCP Client Tool** node.
- Change **Tools to Include** from All to **Selected**, then choose only company_overview and list_competitors.
- Save, then run prompt 4 again. The agent can no longer fetch pricing or news, so the report is thinner. This shows how you control what an agent is allowed to do.
- Switch back to **All** afterwards and save.

8. Troubleshooting

Agent says it has no tools, or the tool list is empty	Open the MCP Client Tool node and check Server Transport is HTTP Streamable (the default is SSE). Check the endpoint ends in /mcp and starts with https. Save the workflow and ask again.
MCP Client Tool shows a connection or timeout error	Check Terminal 1 (server) and Terminal 2 (ngrok) are both running. Open <your ngrok url>/health in a browser. If ngrok was restarted, copy the new Forwarding URL and update the Endpoint.
No new lines appear in Terminal 1	The request is not reaching your laptop. The endpoint in n8n is probably an old ngrok address. Copy the current one and paste it in.
ngrok says ERR_NGROK_4018 or authentication failed	Run ngrok config add-authtoken <token> with the token from dashboard.ngrok.com/get-started/your-authtoken, then start ngrok again.
Gemini credential error, or API key not valid	Re-copy the key from aistudio.google.com/app/apikey with no spaces. If it still fails, create a new key in a new project and update the credential.
Quota, rate limit or 429 error from Gemini	Wait a minute and try again. Choose a Flash model, use a shorter prompt, or check your limits in Google AI Studio.
Agent answers from its own knowledge and does not call tools	Check the system message was saved (Step C8) and that the MCP Client Tool is connected under the Tool connector of the agent. Mention MarketIntel in your prompt.
Results say “DEMO MODE: no canned data for ...”	Demo mode only knows NVIDIA, AMD and Intel. Use one of them, or add your Tavily key to .env, restart the server and try again.
Agent keeps repeating earlier answers	Simple Memory is carrying over the earlier conversation. Start a new chat session and try again.
Port 8000 is already in use	Set PORT=8010 in the .env file, restart the server, run ngrok http 8010, and update the Endpoint in the MCP Client Tool with the new ngrok address.
n8n will not run the workflow, or a limit message appears	Check your n8n Cloud plan or trial for remaining executions.

9. Cleanup and safety

- The MarketIntel server has no login. Anyone who knows your ngrok URL can call it, so

stop ngrok (Ctrl+C in Terminal 2) when the demo is over.

- Stop the server too (Ctrl+C in Terminal 1).
- Treat the Gemini key and the ngrok authtoken like passwords. Do not show them on screen. Delete the Gemini key in Google AI Studio after the training if you no longer need it.
- All MarketIntel tools only read information. They do not write or delete anything.
- If you later want to protect the server, n8n's MCP Client Tool supports Bearer and Header authentication, and the server can check a token.

10. Quick reference

Start the server	uv run server.py (from the MarketIntel folder)
Start the tunnel	ngrok http 8000
MCP URL for n8n	https://<your-id>.ngrok-free.app/mcp
Health check	https://<your-id>.ngrok-free.app/health
MCP Client Tool settings	Transport: HTTP Streamable | Authentication: None | Tools to Include: All
Model	Google Gemini Chat Model (a Flash model)
Gemini key page	aistudio.google.com/app/apikey
Agent name	Research Agent